Linear: one weighted score per class

The simplest baseline connects all 784 pixels directly to ten output scores.

Flat pixels
784 values

→

Class weights
10 sets of 784

→

Weighted sum
z = Wx + b

→

Prediction
10 logits, argmax

How one score is calculated

For each class, multiply every pixel by its learned weight, add the products, then add a bias.

W has shape 10 × 784. The bias has 10 values: 7,840 + 10 = 7,850 parameters.

What this baseline can learn

Each class learns a position-specific pixel pattern. There is no hidden layer or nonlinear feature extractor.

All models train on raw logits with cross-entropy. Softmax is unnecessary before the loss or argmax.

7,850 parameters	Accuracy 84.26%	Macro-F1 0.8417	Selected epoch 9

Shapes describe one image. Scores and epochs: original PDF / comparison.json, seed 42.

MLP: nonlinear combinations of pixels

Two hidden layers combine the flat input into increasingly abstract features.

Flat pixels
784 values

→

Hidden layer 1
256 units

→

Hidden layer 2
128 units

→

Class scores
10 logits

Inside each hidden layer

A dense layer mixes all its inputs. ReLU keeps positive activations and sets negative ones to zero.

Dropout 0.2 follows each ReLU during training. Validation and testing disable dropout.

What changes from Linear

Nonlinear hidden layers can combine pixel evidence in ways a single linear map cannot.

The MLP keeps every pixel, but uses no shared local filters. Its 4.24-point accuracy gain also comes with greater capacity.

235,146 parameters	Accuracy 88.50%	Macro-F1 0.8846	Selected epoch 8

Shapes describe one image. Scores and epochs: original PDF / comparison.json, seed 42.

CNN: shared filters learn local patterns

Convolution keeps the image grid, while pooling reduces its spatial size.

Image grid	→	Conv + pool	→	Conv + pool	→	Flatten + dense	→	Class scores
1 × 28 × 28		32 × 14 × 14		64 × 7 × 7		3,136 to 128		10 logits

Inside the two convolution blocks

Each block uses a 3 × 3 convolution, padding 1, ReLU, then 2 × 2 max pooling.

Convolution preserves spatial size. Pooling halves it: 28 to 14 to 7. Shared filters reuse a detector at different locations.

The classifier holds most weights

Flatten 64 × 7 × 7 features, then dense 128, ReLU, dropout 0.2, and dense 10.

The first dense layer has 401,536 parameters, about 95% of the total. The best score here does not isolate a pure convolution effect.

421,642 parameters	Accuracy 91.94%	Macro-F1 0.9189	Selected epoch 10

Shapes describe one image. Scores and epochs: original PDF / comparison.json, seed 42.

LSTM: a memory updated one row at a time

The 28 image rows form a spatial sequence, read from top to bottom.

Row sequence
28 steps × 28 pixels

→

LSTM layer
128 hidden units

→

Final hidden state
128 values, h at row 28

→

Readout
Dropout 0.2 + dense 10

How the memory changes

At each row, the forget gate controls old memory, the input gate controls new content, and the output gate exposes the hidden state.

Hidden and cell states start at zero for each image. The same LSTM weights process every row.

Exactly which state is classified

A single forward layer reads all 28 rows. Only its final hidden state enters the classifier.

The cell state carries memory internally. There is no averaging of row outputs. The 88.48% score is only two test images behind MLP.

82,186 parameters	Accuracy 88.48%	Macro-F1 0.8835	Selected epoch 9

Shapes describe one image. Scores and epochs: original PDF / comparison.json, seed 42.

Transformer: patches exchange information

Self-attention connects all 49 patches, using learned positions to retain layout.

Patch sequence
49 patches × 16 pixels

→

Embed + position
49 tokens × 64

→

2 encoder blocks
4 heads per block

→

Normalize + mean
49 tokens become 64

→

Readout
Dropout + dense 10

Inside each encoder block

Each attention head compares queries and keys, then mixes values from all patches. Each head uses 16 features.

Pre-norm residual sublayers wrap attention and a 64-to-128-to-64 feedforward network with GELU. Dropout is 0.2.

From an image to one prediction

Split the image into a 7 × 7 grid of 4 × 4 patches in raster order. Project 16 pixels to 64 features, add positions, then apply dropout.

After the blocks, layer-normalize and average all tokens. No CLS token or causal mask. Training starts from scratch.

71,946 parameters	Accuracy 85.55%	Macro-F1 0.8524	Selected epoch 10

Shapes describe one image. Scores and epochs: original PDF / comparison.json, seed 42.